SYSC3110

UNO Flip – Data Structure Explanation (M3)

ENUM CLASSES

CardColor

- Enum for all card colours (light: RED, BLUE, YELLOW, GREEN, WILD; dark: ORANGE, PINK, PURPLE, TEAL).

- Field: boolean isLight.

- Used in PileSystem to generate light-side cards only.

CardType

- Enum for all card types (light and dark sides).

- Field: boolean isLight.

- Used to differentiate light/dark side behaviour when generating deck or flipping.

Status

- Enum representing the game's current state.

- Used by UNOModel to set state and notify views via UNOEvent.

CARD CLASS

Data Structure: HashMap Point_System

- Purpose: Maps card type to its score.

- Time complexity: O(1) lookup.

- Reason: Fast and clean mapping of type $\rightarrow$ score.

Other Fields:

- Dual state fields (light/dark colors and types).

- ImageIcon light/dark for each side.

- Behaviour depends on UNOModel.isDarkSide.

PILE SYSTEM

Data Structures: Stack drawPile, Stack discardPile

- Purpose: Represent draw and discard piles.

- Operations: push/pop/peek – O(1); shuffle – O(n); restore – O(n).

- Reason: Stack fits LIFO pile behaviour in UNO.

PLAYER CLASS

Data Structure: List hand = new ArrayList<>();

- Purpose: Player hand (cards held).
- Operations: add O(1), remove O(n), get O(1).
- Reason: Hands are small; ArrayList gives simple index access.

UNO MODEL

Data Structures:

- ArrayList players
- List viewersList
- PileSystem pile
- String[] playerOptionsLight/Dark

Purposes:

- players: hold all players (humans and computers).
- viewersList: observer list for UI updates.
- pile: manages draw/discard via PileSystem.
- arrays: static wild color options.

Complexities:

- Access O(1), iteration O(n).

UNO EVENT

Data Structure: ArrayList players

- Purpose: Snapshot of model state for the view.
- Passed via notifyViewers().
- Complexity: mostly O(1) field access.

UNO VIEW

Data Structures:

- ArrayList playerScoreLabels

- List (from model) for current player’s hand

Purpose:

- playerScoreLabels updates scoreboard quickly.
- refreshCenterCards() rebuilds buttons for each card.

UNO CONTROLLER

- No collection structures.
- Pure mediator between View and Model using ActionListener.

TOP PANEL

- No java.util structures.
- Uses JLabels only for top card, player, status, wild color.

SUMMARY

Core Collections:

- HashMap → scoring.
- Stack → draw/discard piles.
- ArrayList → player hands.
- ArrayList → game participants.
- ArrayList → observers.
- ArrayList → scoreboard UI.

Most operations: O(1) to O(n) where n is small.

Design keeps it simple, efficient, and aligned with UNO gameplay.